Large Language Models

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Plan for today

1. Recap: Transformers
2. Large Language Models
3. DeepSeek

Recap: Transformers

→ Blackboard.

Language Modeling

Predict the next word.

Estimate $p(x_t | x_{t-1}, x_{t-2}, \ldots, x_0)$

By adding a linear *language modeling head* on top of the transformer. The language modeling head is just a linear layer with $N_{\text{vocabulary}}$ outputs, followed by softmax & cross-entropy loss

Large Language Models

Three main training stages

1. Pre-training
2. Supervised Finetuning
3. Reinforcement learning

Pretraining

Language modeling objective

Large amounts of low-to-medium-quality data

Supervised Finetuning (SFT)

Language modeling objective

Small amounts of high-quality data

Reinforcement Learning from Human Feedback (RLHF)

1. Language model generate utterances
2. Humans judge the responses
3. Reward model is trained to "simulate" human judges
4. Language model is updated to increase reward

LLM Paperstorm

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Goal: Investigate the **model architectures** of recent LLMs.

Paper pool:

- Gemma 4
- DeepSeek V3.2
- Qwen3

What to do?

- Groups of 2-3 people.
- 1 paper per group.
- 20 Minutes, then we bring together results.
- **Hint**: You may need to look into previous papers!
- **Tools**: You may use AI assistance as you desire.